

UNIVERSITY OF OULU

FACULTY OF INFORMATION TECHNOLOGY AND ELECTRICAL ENGINEERING

Program Committee

TODO: BACHELOR'S AND DIPLOMA THESIS GUIDE

Master's thesis
Degree Programme in Computer Science and Engineering
June 2026

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ABSTRACT

This document provides general guidance for a degree student in the technical degree programmes of Computer Science and Engineering, Biomedical Engineering and Business Analytics (Tech.) at the faculty of Information Technology and Electrical Engineering at the University of Oulu in preparing his/her Bachelor's or Master's thesis. This guide defines the role of a thesis in the M.Sc. (Tech.) Degree, presents the actions to be taken in different phases of the thesis procedure and introduces the way that the thesis is written. The document has been formatted based on these guidelines to serve as an example how the thesis should look like.

Keywords: M.Sc. degree, master's thesis instructions, structure of a master's thesis[1]

[1]The keywords should be different from the ones used in the title. Remove this footnote from your thesis.

Committee P. (2026) TODO: Kandidaatintyön ja diplomityön teko-ohjeet. Oulun yliopisto, Tieto- ja sähkötekniikan tiedekunta, Tietotekniikan tutkinto-ohjelma, Diplomityö, 41 s.

TIIVISTELMÄ

Näissä ohjeissa opastetaan valmistumisvaiheessa olevaa opiskelijaa kandidaatin- ja diplomityön tekemisessä. Ohjeissa selvitetään työn asema diplomi-insinöörityötkinnossa, kerrotaan toimenpiteet, joihin työn tekijän on ryhdyttävä työn eri vaiheissa, sekä määritellään yksityiskohtaisesti kandidaatin- ja diplomitöiden kirjallinen rakenne. Ohje on muotoiltu noudattaen näitä kirjoitusohjeita ja toimii siten esimerkkinä, miltä lopputyön tulisi näyttää.

Avainsanat: diplomi-insinöörin tutkinto, diplomityön ohjeet, diplomityön rakenne[2]

[2]Käytä avainsanoina eri sanoja kuin lopputyön otsikossa. Poista tämä alaviite lopputyöstäsi.

TABLE OF CONTENTS

ABSTRACT
TIIVISTELMÄ
TABLE OF CONTENTS
FOREWORD
LIST OF ABBREVIATIONS AND SYMBOLS
1. INTRODUCTION 9
    1.1. Thesis Guide 9
    1.2. Author's Contributions and the Role of Artificial Intelligence 9
2. GENERAL OUTLINE FOR THESIS PROJECTS 11
    2.1. Approval of the Topic 11
    2.2. Writing the Thesis 12
    2.3. Delivering the Thesis and Applying for the Degree 13
    2.4. Evaluation and Grading 14
    2.5. Publicity 14
    2.6. Thesis Awards 15
3. THESIS WRITING INSTRUCTIONS 16
    3.1. Linguistic Style 16
    3.2. Typography 17
        3.2.1. Figures and Tables 17
        3.2.2. Algorithms 20
    3.3. Clarifying Contributions and Using AI 20
    3.4. Practical Advice 21
        3.4.1. Do Not Leave the Writing to the End 21
        3.4.2. Discuss in Detail with Your Supervisor Before You Start Writing and Design a Body for Your Thesis 21
        3.4.3. Write Things in the Right Order 22
        3.4.4. Write Simply 22
        3.4.5. Ask Your Technical Supervisor to Go through What You Have Written 22
        3.4.6. Do Not Get Stuck 23
        3.4.7. Hints for Editing 23
    3.5. Electronic Version 24
    3.6. Optional Bound Versions 24
    3.7. Copyright, Licenses and the Use of Images 25
    3.8. Accessibility 27
4. THE LITERARY STRUCTURE OF A THESIS 28
    4.1. First Pages 28
        4.1.1. Title Page 28
        4.1.2. Abstract 28
        4.1.3. Abstract in Finnish 29
        4.1.4. Table of Contents 29
        4.1.5. Foreword 29
        4.1.6. List of Abbreviations and Symbols 29
    4.2. Introduction 29
    4.3. Core Text Part of Your Thesis 30

4.4. Discussion 32
4.5. Summary 33
4.6. References 33
4.7. Appendices 35
5. SUMMARY 36
6. REFERENCES 37
7. APPENDICES 40

FOREWORD

These guidelines are based on thesis instructions of the former Department of Electrical and Information Engineering, the instructions of the publication series Acta Universitatis Fennica, and the book "Teknisen kirjoituksen laatiminen" [1–3]. Several persons from the former Department of Electrical and Information Engineering have participated in setting up these instructions. The preparation of the first version was led by Prof. Pentti Lappalainen.

The Finnish guidelines were edited by the Study Committee of the Department of Electrical and Information Engineering in 2005 and 2010-2011, and revised by the Degree Programme Committee of Computer Science and Engineering in 2011 and 2012. The original English version was written by the Degree Programme Committee of Electrical Engineering; this version is based on that version and the Finnish instructions for Computer Science and Engineering. This English version has been updated to comply with the new conventions for archiving the electronic versions of master's theses in 2013, and the organization renewal in 2016 where the Department of Computer Science and Engineering was disbanded and divided into three research units.

The instructions were revised in July 2018 for the Faculty of Information Technology and Electrical Engineering. The update was coupled with the transition to Overleaf as recommended LaTeX document preparation platform. The key modification was the overhaul of referencing to IEEE style. The revisions in 2023 were made by the Computer Engineering Degree Program Committee to improve instructions and to clarify the use of footnotes and references in case of pictures.

The latest revision was done in August 2025. This included a complete overhaul of the LaTeX template. Stylistic details have been unified with with the guidelines of the Degree Programme of Electronics and Communication Engineering, though some differences remain. The instructions have also improved the instructions on copyright issues, especially involving images and proper citation style.

This document is applicable for Bachelor's and Diploma theses starting or completing in and after September 2025.

Oulu, 5th June, 2026

Program Committee

LIST OF ABBREVIATIONS AND SYMBOLS

CSE[3]	Computer Science and Engineering
GUI	graphical user interface
LBP	local binary pattern
NCC	normalized cross-correlation
SVM	support vector machine
UML	unified modeling language
f	focal length
N	number of samples
$\mathbf{Q}$	orthogonal matrix
r_{ij}	matrix element
s	distance
t	time
v	velocity
$\mathbf{v}$	velocity vector
$\mathbf{x}$	planar coordinate
β	shape factor
ϵ_{k}	error signal at instant k
$\mathbf{\Phi}_{\mathrm{k}}$	state transition matrix at instant k
$\boldsymbol{\theta}$	parameter vector
σ^2	variance
$\lfloor\ \rfloor$	integer part
$\oplus$	logical XOR operation
arg()	argument

[3]Even though a symbol or an abbreviation is explained on this page, it should be written out full when appearing in the text for the first time. Remove this footnote from your thesis.

1. INTRODUCTION

Writing a thesis takes on a major part of completing the master's degree studies. The thesis assignment prepares for the independent engineering work. Hence, supervision plays a smaller role in thesis procedure than during previous studies. A typical master's thesis represents a solution to a relatively extensive technical problem. Additional studies in the given field are often necessary; however, the aim of the thesis work is to make use of the knowledge and skills acquired during preceding studies. Furthermore, technical and scientific documentation skills will be strengthened. The thesis work can be a conducted as a part of a larger project, but the master's thesis itself should be written individually.

The thesis work is undertaken in the final phase of the studies. It is recommended to begin the thesis work during the autumn term of the 2nd study year in the master's programme or the 5th year after starting in the bachelor's programme. However, timing is flexible and it is also possible to begin earlier depending on advancement in studies. As a general rule, it is time to get started, when there are 15 to 30 credit points left of the total coursework (in addition to the master's thesis). Some fields of study require certain courses to be completed before the master's thesis. Requirements should always be checked in advance with the supervisor. Information on the degree specific requirements is provided by the secretary of the degree program and the registered credit points can be viewed in Peppi.

1.1. Thesis Guide

The aim of these instructions is to give detailed guidelines for composing and writing a master's thesis. This document describes the thesis work process; starting from searching the topic to the formal approval of the completed thesis. The process has otherwise stayed the same for quite a while, but the final stages were changed in the beginning of 2013 due to electronic archiving for all master's theses at the University of Oulu. Hence, the older versions of this document are no longer valid and should not be used. Moreover, this document offers practical instructions for the writing of the thesis: for the literary structure, the layout, and for the writing process as well. This document itself has been edited according to the instructed layout, though the literary structure is not similar to that of a master's thesis.

Before starting your master's thesis work, **please first read this document carefully.** Following these instructions closely is likely to produce a better result and a higher grade. When the supervisor does not have to point out about these instructions (specifically about the layout) she/he can focus on instructing on the content and how the best grades can be achieved. The degree programme web pages [4] offer instructions for master thesis as well. When neither of these sources gives an answer to your questions, please ask your supervisor or from the study affairs office. And remember: Although writing the thesis can be difficult at times, every M.Sc. graduated from Faculty of ITEE has succeeded in this effort!

1.2. Author's Contributions and the Role of Artificial Intelligence

The structure of a thesis can vary. However, at the end of the introductory chapter, you must include a section titled *Author's Contributions and the Role of Artificial Intelligence*. In this

section, you should clarify your independent contributions to the thesis and explain how AI was used in its preparation, if applicable. More information can be found in Sections 2.2 and 3.3.

2. GENERAL OUTLINE FOR THESIS PROJECTS

The process starts with specifying the topic of the thesis and applying for its formal approval by the degree programme committee head. You need to define the topic together with the commissioner and the supervisors (see the next section). The topic, commissioner, and supervisors are specified in the topic application form that is used to apply the approval of the topic. The list below presents the general thesis process, with emphasis on the administrative issues and supervision:

1. approval for the topic,
2. writing the thesis (preferably at the same time as carrying out the research),
3. delivering the thesis and applying for the degree,
4. evaluation and grading, and
5. granting the degree.

The second stage is the most laborious part, which includes writing the thesis and performing the actual work, for example, designing, building and testing a piece of software. When the thesis is ready, it is reviewed by the supervisors, modified by you if necessary, and then delivered to the Laturi system. The next step is to apply for the degree certificate and to perform the other tasks that are required from graduating students. The supervisors download the thesis from the archive and evaluate it. The degree programme committee approves the grading in its meeting and the faculty grants the degree. The thesis is transferred to the university archive and published according to the publicity level you defined when uploading the thesis to Laturi. The graduation ceremony ends your M.Sc. studies. The process is explained in more detail in the rest of this document and on the degree programme web pages [4].

2.1. Approval of the Topic

Your first task is to find a commissioner for your thesis that is typically a company which hires you to make a thesis project. For example, a summer job or training in a company or some other organization gives a good starting point to continue in a thesis project. A research unit in the Faculty of Information Technology and Electrical Engineering can also act as the commissioner, in which case the topic will most likely be linked with ongoing research at the unit. Research units provide information about current master's thesis topics in their web pages. You can also directly ask topics from the research unit personnel. In some cases the research units can pay salary to the master's thesis worker, but in most cases you need to be prepared to make the thesis for free.

There are 2-3 supervisors involved in the master's thesis process, and they are assigned after the topic is confirmed. *Principal supervisor* is a professor or a doctor who belongs to the personnel of some ITEE research unit. A list of potential supervisors and their fields of expertise can be found on the degree program web pages [4]. The professor or doctor you approach first can decide based on the thesis topic to act as a principal supervisor her/himself or suggest someone else. Usually, the thesis is done on a topic within the area of your orientation (major). However, topics proposed by companies are often multi-disciplinary or cross-scientific and the topic does not fit into the realm of any particular orientation. In these cases, the supervision

should be agreed with a professor or doctor who best represents the overall field of the thesis work.

If the master's thesis is made to a university research unit, the commissioner often becomes the principal supervisor. If the thesis is made to a company or some other organization outside the university, the commissioner should assign a *technical supervisor* who is responsible for advising you in the technical matters. It is also common to have a technical supervisor for theses made to university research units. In that case the role of the principal supervisor is mainly to ensure that the requirements set for a master's thesis are fulfilled. In addition, there is always a *second examiner* from the university who is selected by the principal supervisor. The second examiner evaluates the thesis together with the principal supervisor and sometimes also participates to supervision in the issues related to her/his expertise.

If the thesis work is conducted outside the university, it is important to find the principal supervisor as soon as possible in order to make sure that the topic is suitable and to agree on the scope of the thesis. It is recommended to organize a meeting together with the commissioner and the principal supervisor to discuss about the thesis work before it has been started. It is also recommended to write a short project plan that describes the background, motivation, and objectives of the work as well as a timetable and possible risks. The project plan gives a good basis for following the progress of the work.

After the supervisors have been selected, invite the main supervisor in Laturi. The head of the degree programme will will formally appoint the role. After the aims have been agreed with the supervisors, you are ready to submit the research plan for your thesis. You should apply for approval as soon as possible, to be able to take the possible feedback into account in your thesis work.

As a rule, Bachelor's thesis is written in Finnish. It can be written in English in exceptional cases, for example, if the supervisor is English-speaking or if the thesis project group includes members who do not speak Finnish. In English-language degree programmes the Bachelor's thesis is written in English. For Master's thesis, the student can choose the language. If the work is written in English, proofreading by a professional translator can be required. The decision is made by the principal supervisor. The author is responsible for paying the proofreading costs unless it is specified otherwise in the application. It should also be noticed that language is one of the thesis evaluation criteria.

2.2. Writing the Thesis

During the actual work, you are expected to communicate regularly with the technical and principal supervisors. In the beginning, meetings with the principal supervisor can be less frequent, but during the write-up phase, regular meetings are more important.

Every master's thesis is a unique project and it is difficult to define general rules what the individual steps are and how frequently the meetings with the supervisors should be arranged. The Master's thesis gives an opportunity to demonstrate your professional maturity, but it is also the final learning possibility at the university. Hence, it is important to find the balance between the independent work and the amount of supervision required, which is one of the thesis evaluation criteria. You are entitled to get the necessary supervision, but do not expect that the supervisor will make the thesis for you. During the actual work it is recommended to have frequent appointments with the technical supervisor for discussing about the issues that help

you to solve the problems encountered in the work. These meetings also keep the supervisor updated and ensure that you are not stuck or going to wrong direction.

At the beginning of the write-up phase, guidance from the principal supervisor is crucial concerning the structure, presentation order and style of the thesis. During the write-up, meetings with the principal supervisor should be held in order to discuss a) whether the information order and emphases are right, b) whether the issues being covered or planned to be covered are relevant to the thesis, and c) whether some areas have been overlooked. The emphasis in the meetings with the principal supervisor should mainly concern the structuring of the thesis.

At minimum, you should present the table of contents to the principal supervisor before you start writing the thesis, present the first complete draft when it is ready, and in this meeting agree the necessary modifications, and then review the modifications in follow-up meetings. In these final meetings (one or more), special attention needs to be paid to correct referencing of others' work and to the rights and permissions to the content included in the thesis. When the supervisor judges the thesis to be ready, she/he permits electronic publication of the thesis.

Further, always discuss the use of Artificial Intelligence-powered tools with your supervisor(s) well before writing the thesis. You must include a dedicated subsection at the end of the Inroduction chapter, titled *Author's Contributions and the Role of Artificial Intelligence*. The statement must first clearly clarify what was the role of the thesis author in the included work and, second, how AI was used in preparing the thesis. Please familiarise yourself with the more specific guidelines in Section '*Clarifying contributions and using AI*' in Chapter 3 of this guide.

2.3. Delivering the Thesis and Applying for the Degree

Starting from January 2013, master's thesis have been published only in an electronic form. When the thesis is ready and the supervisor has given the permission, the student delivers an electronic version of the thesis to the Laturi system[4]. Since the supervisors fetch your thesis for evaluation from Laturi, MSc students need to deliver the thesis no later than 10 days prior to a degree programme committee meeting approving your thesis. You can discuss the details of the schedule with the supervisor. Committee meetings are held mainly once per month, and the meeting schedule can be found from the degree programme web pages (for BSc students [5] and MSc students [4]). After submission to Laturi, BSc students will send the final version of the thesis to the study office as well.

The delivered version must be the final version of the thesis manuscript[5]. The thesis must have the official title page and the layout and general structure presented in this document. **Pay special attention to the title page, as you have to set your degree program correctly.** The new template can be downloaded from degree programme web pages [4]; old templates are no longer valid.

Once the thesis is accepted by the supervisors and graded by the degree programme committee, it is transferred to the university archive and published at the OuluREPO repository[6] or e-thesis workstations in the university library according to the publicity level you determined when you delivered the thesis.

[4]http://laturi.oulu.fi/. A good name for your thesis file is <year>-<month>-<familyname>-<firstname>-Thesis.pdf. When several theses are downloaded for review from Laturi, this naming convention helps to identify the files

[5]A second version of the thesis can be uploaded to the Laturi system only for a cogent reason and requires a permission from the university library (this permission has to be applied by the student).

[6]https://oulurepo.oulu.fi/

The application for a diploma is submitted via Peppi's graduation service at student's desktop. In addition, there is a feedback questionnaire for BSc students. MSc students can download TEK Survey for Graduates from [6]. The secretaries will prepare your papers, calculate your GPA, etc.

Before graduation you also need to attend the **maturity test.** Passing the maturity test is required for all degree students and it is taken after the completion of the thesis. The students with Finnish/Swedish as the language of their elementary education will write the test in Finnish/Swedish. If the elementary education has been taken in some other language, the maturity test is taken in English. The maturity test is a written examination based on the thesis, where the candidate is asked to write an essay about the topic(s). The registration for the maturity test should be discussed with the supervisor as you are getting your thesis approved. Finally, you need to return all books, keys, equipment, machinery, and tools that belong to the university.

2.4. Evaluation and Grading

The principal supervisor and the second examiner download the thesis from Laturi for **final evaluation and grading.** They fill out the evaluation form with the proposed grading in Laturi. If the thesis was commissioned by a company or an organization outside the university, the technical supervisor is expected to fill out a separate evaluation form and send it to the principal supervisor well in advance before the grading. The degree programme committee approves the thesis based on the evaluation by the principal supervisor and the second examiner. The forms and a document describing the evaluation criteria can be found from the degree programme web pages [4].

A master's thesis that has been approved will be graded according to the university regulations with the following 5-point scale: satisfactory (1), very satisfactory (2), good (3), very good (4), and excellent (5). The title of the thesis, the name of the Degree programme director, as well as the grade will be printed on the Master's Degree Certificate.

The student has the right to see the grade proposed by the supervisor, as well as the evaluation form, **three** days prior to the committee meeting. In case the student feels mistreated, she/he has the opportunity to issue a written **appeal** to the board of examiners of the University of Oulu concerning the evaluation of his/her thesis no later than **fourteen days** after having received the information. This process evidently delays the graduation.

2.5. Publicity

A master's thesis is a public document. A thesis must therefore not reveal any business secrets or confidential information. The thesis can only be evaluated based on its written contents. For this reason, it is crucial to discuss thesis publicity with company representatives as early as possible, when the topic is defined. Should any major conflicting interests arise between the author and the customer concerning the publication of information, the author should turn to the principal supervisor for consultation.

The Ministry of Education has issued a set of written statements concerning the public nature of master's theses to universities and colleges. According to the statements the thesis must not contain classified information, and once approved, the thesis should be public. The ownership as

well as the publication and/or patent rights should be agreed upon together with the supervisor, author, and the possible commissioner.

Starting from January 2013, all master's thesis written at the University of Oulu have been published electronically by the university library. The publicity level is decided by the author when the thesis is delivered to Laturi; whether the electronic version will be available to everyone from an open access repository or only at special workstations in university libraries.

2.6. Thesis Awards

The Technological Society of Finland and the IEEE Finland section issue the Best Thesis of the Year Award, chosen from a list of candidates that is compiled by Finnish universities and colleges and is including almost all theses published annually in Finland.

3. THESIS WRITING INSTRUCTIONS

A Master's thesis in engineering usually consists of an implementation part (literature survey, hardware development, software development, measurements, etc.), and a written part (the text body). How the implementation part is carried out depends on the topic, so no general guidelines can be issued. However, each type of publication has its distinct layout and structure to be followed. The following set of instructions will explain in detail the writing and layout style for the master's theses published in the engineering degree programmes at the Faculty of ITEE. Typography affects the readability of the text greatly, so the instructions should be followed strictly. By following these instructions, the thesis authors can properly learn a proper way of expressing themselves formally in writing. After having learned one way of formal writing, it is easier to get used to the formalism used at your future workplace. Presentation and layout style have an effect on the grade of the thesis.

Before starting to write your thesis, it is a good idea to mind-map or chart out the various issues that will be included in a thesis. After that, you should divide them into themes, actual chapters with actual title, and estimated numbers of pages to be included in each chapter. The contents should be discussed in detail with your principal supervisor. Pay attention on the weighting and focus of different topics. You should reserve enough time for writing your thesis, so that its content and structure will be as good as possible. It should be remembered that the archived thesis book is usually the only document through which your whole thesis work will later be assessed.

3.1. Linguistic Style

As a general rule, students who have completed their matriculation exam in Finland write their Master's thesis in Finnish or Swedish. Foreign students who are not fluent in Finnish write their thesis in English. The thesis can be also written in English if it is well-justified, for example, when the supervisor or the project team does not understand Finnish well enough.

The language of a thesis written in English has to be flawless. If the thesis does not meet this requirement, or the principal supervisor feels that she/he is not capable of checking the language well enough, the supervisor can demand the thesis to be checked by a proofreading company, a person qualified to edit technological vocabulary in English, a native English speaker, or a person holding the degree of M.A. in English Philology. One should notice that the correctness of the language affect the grading of the thesis.

The thesis should have an abstract and a title in both English and Finnish. International students **also require a Finnish abstract and title** (your supervisor can help with getting a native speaker to verify these). If the language of the thesis is English, the abstract in English comes before the abstract (tiivistelmä) in Finnish.

A Master's thesis is written for an engineering audience. The author should therefore avoid presenting issues and topics that go beyond this scope. You should apply professional terminology when available. This rule also applies to all figures and tables.

The aim is a clear and well-structured thesis, **without unnecessary excessive use of words**—a thesis usually has 50–80 pages. The language (English/Finnish/ Swedish) should be fluent and readable, and it should adhere to the conventions and recommendations applied in a particular language. Advice on such issues can be obtained from various language guides (in Finnish,

e.g., [7]). There are also several excellent sources on the internet, e.g., [8, 9] in Finnish and [10, 11] in English.

3.2. Typography

In the text typography, you need to use the following guidelines and rules:

- Font: Times New Roman
- Page margin settings (exception: physical, bound copies use different settings; see Section 3.6):
  - Left margin: 2,5 cm
  - Right margin: 2,5 cm
  - Top margin: 2,5 cm
  - Bottom margin: 3,0 cm
- Spacing:
  - Before a heading: 2 empty rows
  - After a heading: 1 empty row
  - Between two headings: 1 empty row
  - Line spacing: the default value for each font size, which is usually the font size 2 pts.
  - The space between two paragraphs is a normal line spacing
  - There is no indent in the first paragraph following a heading, but the following paragraphs are indented by 0.5 cm
  - A single empty line is added between table and figure captions and the basic text

3.2.1. Figures and Tables

The thesis should include figures and tables to support the text. They should be placed in the following manner as is automatically done when following the model of this LaTeX template:

- Table structure and the different fonts used in different instances are explained in Table 1.
- In tables, the heading has to be placed above the table. The table heading should not end in a full stop. The permission to publish a table copied from another source has to be mentioned as part of the table heading. Many publishers inform the correct form to write this permission when such a permission has been granted. The table must be in text format – not an image.
- If necessary, the caption should be separated from the figure with a single line spacing for reasons of clarity. A figure and a short caption are centered. However, a caption spanning multiple lines is justified at both ends. The necessary references are included in the figure captions. Copyright information is also added to the end of the caption. The permission to publish a figure copied form another source has to be mentioned as part of the figure text. Many publishers inform the correct form to write this permission when such a permission has been granted. The copyright issues and citation practices of images are further discussed in Section 3.7.
- Do not start a chapter with a figure, but embed it in the text content. A figure is supposed to always appear in the text after its reference. If there is not enough space for the figure

directly after the first reference, the word processor moves the figure automatically to the next page. In such case, the figure can be moved forward so that text originally after the figure fills the empty space. However, a figure should always be placed in the same chapter—hence sometimes empty space at the bottom of some chapters cannot be avoided (the same rules applies for the tables).

Table 1: The font types used in a Master's thesis

Font size (pt)	Layout			
	Regular	**Bold**	*Italic*	***Bold Italic***
10	Table and figure contents, footnotes, and endnotes			
12	Standard text equations, references, tables, captions, table headings	**1st level subheadings, abstract, abstract in Finnish (tiivistelmä)**	*3rd level subheadings*	***2nd level sub-headings***
14		**CHAPTER TITLE**[1]		
16		**Author name**		
18		**THESIS TITLE**		

[1] Each full chapter should start on a new page.

You should not use single subsections—for example, Section 2.1 needs to be accompanied by Section 2.2. Also, please do not number sections beyond the 3rd grade (e.g., 1.2.1.1 should not be used). Should the need arise for further decimalization the following method below (a **bolded**, unnumbered heading that does not appear in the table of contents) should be applied:

Formatting the figure captions

Figures, tables and appendices are a part of the written presentation. All these need to be referenced to in the text body, preferably before the figure is placed in the text—i.e., first the referring text, then the figure or table. Figures and tables have a running number through the document—or chapter wise, if there are plenty of figures. Figure 1 serves as an example of a figure and a text referring to it. Figure captions are below the figure, and the caption text ends with a full stop. A short caption is centered, while a long caption extending to a several lines is justified on both sides.

If no external source is mentioned in the caption, it is assumed that the thesis author is also the creator and copyright holder of the figure (noting that authorship and copyright ownership

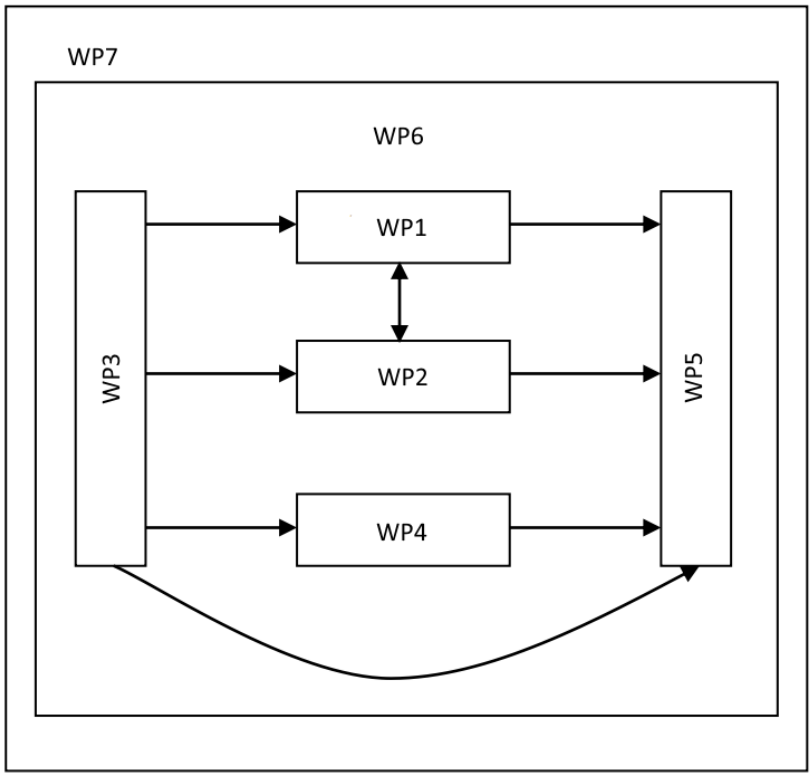


Figure 1: Connections between work packages (WP). Copyright © 2026 Program Committee. CC BY 4.0.

are not always the same). In such cases, the figure is licensed under the same terms as the document itself (default: "All rights reserved"). This is further discussed in Section 3.7. For figures that constitute significant contributions — and may therefore be of interest for reuse — it is advisable to explicitly state the copyright and licensing information, even if it has already been included on a separate copyright page.

Figures and tables should be given an alternative text that summarises the information they contain for the reader. This is discussed further in Section 3.8. Unfortunately support for this in LaTeX is still work in progress, and the alternative text will currently not end up in the final PDF, so including them is not a requirement. A suitable alternative text for Figure 1 could be:

> Diagram showing interactions between work packages WP1 to WP5, all inside WP6, which is itself inside WP7. WP3 (left) sends outputs to WP1, WP2, and WP4. WP1 and WP2 are bidirectionally linked. WP1, WP2, and WP4 send outputs to WP5 (right). A curved arrow shows WP3 also directly feeding into WP5.

The final thesis will be published as a PDF/A file, and this LaTeX template produces files in that format directly. Be sure to remove all transparency in your source images to ensure the end result is also compliant. This affects figures in PDF and PNG formats. Instructions on doing this are included with the thesis template. If you have done this for all of your images, there is no need to use Muuntaja.

3.2.2. Algorithms

Algorithms can be implemented simply using LaTeX. An example of an algorithm is given in Algorithm 1. An input and an output for the algorithm should be assigned at the start. Equations or discussed methods can be referenced in the algorithm. Depending on one's needs the algorithm may be implemented in a higher level of pseudocode compared to one in Algorithm 1.

Algorithm 1. Selection Sort

```
   Input  : An unsorted list a of size n
   Output : A sorted list a of size n
   :
 1 for i ← 0 to n − 1 do
 2 |   min ← i
 3 |   for j ← i + 1 to n do
 4 |   |   if a[j] < a[min] then
 5 |   |   |   min ← j
 6 |   |   end
 7 |   end
 8 |   if min ≠ i then
 9 |   |   swap (a[i], a[min])
10 |   end
11 end
```

3.3. Clarifying Contributions and Using AI

Since a thesis is an independent work, it is important to clarify how much of the work was done by the author, how much by potential collaborators, and how much was Artificial Intelligence used in the process, if applicable.

Generative AI is useful for various tasks, and its use is expected to increase. Therefore, we are generally positive about exploring the use of AI. However, the responsible and ethical use of AI is critically important. Always discuss AI use with your supervisor prior to writing the thesis, and familiarise yourself with the university's AI policies (https://www.oulu.fi/en/students/general-information-about-studying-at-the-university/guidelines-on-the-use-of-ai-in-education).

As a general rule, do not generate parts of your thesis; presenting content not produced by you as your own is considered fraud (in most cases, plagiarism) and is a serious ethical violation subject to appropriate consequences. You alone are responsible for the accuracy, correctness, and originality of your thesis and its parts. AI cannot be cited as a source of scientific facts and should not be used as a reference. With your supervisor's approval, AI can be used to assist in tasks such as brainstorming, planning, aiding information retrieval, creating or editing visuals (or other modalities), and language proofreading. Any other use cases also require approval from your supervisor. In all cases, it is your responsibility to ensure that the tools you use (including their licenses, terms, and conditions) permit the output to be utilized in your thesis as intended.

The thesis must include a disclaimer at the end of the introduction in a subsection titled "*Author's Contributions and the Role of Artificial Intelligence*." This subsection serves two

purposes. First, you should clarify your own contributions to the thesis, which is necessary in cases when the thesis is completed, for example, in a project with multiple contributors. Second, the section must detail the specific AI tools that were used, the purposes for which they were used, and in which parts of the thesis. For example: *The author of this thesis designed the artifact, conducted all the research, and prepared the thesis. In preparing the thesis, ChatGPT (GPT-4) was used to correct grammar in all chapters. Additionally, Claude 3.5 Sonnet was used to help implement the evaluated artifacts and to describe the code functionality in Chapter 3.*

3.4. Practical Advice

On average, it takes between two to three months of full-time work to write a Master's thesis—around one finalized page per day. You might be able to write a thesis in a shorter time, but you should understand that **it takes much longer than you think to edit and re-edit a thesis, considering both structure and presentation style.** In the following we have listed some practical advice, which serves the purpose of making it easier to start the writing phase.

3.4.1. Do Not Leave the Writing to the End

Working on the implementation and written thesis in parallel will force the author to clarify and reformulate ideas, which might often lead to new ideas and save time on editing, re-editing and restructuring later. At the least, you should start gathering and getting acquainted with your literature and charting out your written part into clearly defined units and chapters at an early stage. Once you have done this and you know where you are going, you will have a much more secure feeling of the scope of your thesis.

3.4.2. Discuss in Detail with Your Supervisor Before You Start Writing and Design a Body for Your Thesis

It is in the best interest of the supervisor that the student graduates as fast as possible and is left without wider difficulties. She/he therefore has an excellent motive to give help in time. A typical problem at the beginning stage that can be fixed easily is a content structure, in which theory and practice have been clearly divided into two separate parts. If left unfixed, this may often lead to repetition and problems in combining the parts.

Hence, start your writing by planning a content structure, (table of contents), which will function as your backbone all throughout your work. Usually it is a good idea to construct your chapters so that you first give each heading and subheading a working name even if you do not have a final idea of the exact title at this stage. Each structural unit (heading) should have a few code words that act as the key to each heading. You can also look at it this way: code words convey a message to the reader, a message you as the author want to tell them. Later when you have started doing the actual writing, you should look at these code words and crystallize the message in each section to correspond to these code words. Thinking ahead like this will reduce the pain of creation, give you confidence and help you feel that your thesis is solid.

3.4.3. Write Things in the Right Order

Most people find it convenient to write their thesis in the order of their table of contents. It is usually a good idea to start by writing the introduction, because the introduction spells out and defines the aims and scope of the thesis. The other chapters in the opening part of the thesis usually lay out the parameters and working environment, the needed theoretical basis of the thesis etc.; they can also be written fairly early on in the process. The opening part of the thesis also includes the literature review. It is highly recommended that you keep track of your references and document them while you are writing, because you might forget or lose track of your sources later.

3.4.4. Write Simply

The first sentence of a paragraph should define its contents. The following sentences clarify the issue. This method results in a clear and easily understandable way of presenting, since each paragraph should contain information only on one or two separate issues. Paragraphs structured like that will be easy to cut and paste elsewhere, if structural rearrangements are later required. To avoid fragmentation, it is important to not present the same things again in different chapters. It will be easier for the reader to follow the idea when your thesis structure is logical and its linguistic style is systematic throughout the thesis. Hence, do not go "over the top" and try to impress the reader with too extravagant ways of presenting your ideas. Instead, present your case as you would like others to present their thesis.

3.4.5. Ask Your Technical Supervisor to Go through What You Have Written

Sometimes the principal supervisor and a technical supervisor are two separate persons, especially in the case where the thesis work is done in a company. In this case you should first have your work read by your technical supervisor. After that edit your work according to his/her advice before you bring it for revision to your principal supervisor. The technical supervisor has a better impression of your work as he/she interacts with you on a regular basis, and can therefore go through your work much faster. Your principal supervisor, however, can form a better view of your work when you bring him/her a more finished document. You should make good use of the expertise provided by your technical supervisor at all stages of your research and write-up phases. However, do not feel intimidated to show your work to your principal supervisor at the beginning of your writing phase (see Chapter 2 above). It is your official supervisor who makes the final evaluation together with the second examiner. It is therefore important to listen to his/her comments at an early stage to ensure that you are not surprised or disappointed when seeing the completed evaluation form.

It is very usual to get blind to typos in one's own text, and one also easily assumes some topics to be so self-evident that a reader has difficulties to follow the text. This is most easily found by a colleague who is not so familiar with the topic. Alternatively, you can take a break of a few days to look your text with fresh eyes—or just concentrate very carefully on what you are really saying and what you just know without writing it.

3.4.6. Do Not Get Stuck

Figure 2 shows how difficult the writing can be sometimes. If you feel you are not advancing with your writing, although you feel like you know your topic, there might be something wrong in the way you work. In this case, do not waste time in wondering and fretting; instead seek advice in the instructions above. If this does not help, then usually your supervisor(s) can help you in solving your problem. At the most difficult time it is good to remember that every engineer you see out there has once been in the same situation as you are now, and yet they were still able to graduate.



Figure 2: Writing angst and the pain of creation. “Writing your thesis can sometimes make you run outside and howl at the moon. — It’s already the fourth week and I haven’t been able to write anything else besides my name down.”

3.4.7. Hints for Editing

A well-prepared document template like the current one with pre-defined paragraph formats speeds up writing. The use of automatic numbering helps to minimize manual corrections, when the order of figures, for example, is changed.

Most word processors, LaTeX included, have a spelling checker feature that is recommended to be used. This way your supervisor does not need to spend time on spotting and correcting typos, but can concentrate on the actual contents of the thesis.

3.5. Electronic Version

University of Oulu started archiving electronic versions of M.Sc. theses in January 2013. All theses will be uploaded to the Laturi system and transferred from there to the university archive once accepted. The theses will also be published in the university repository, but each student can her/himself decide the level of publicity (see Section 2.5, "Publicity"). It should be noted that the repository provides a short link to a published thesis. Hence, the student can include the link, for example, in her/his CV, and generally use a publicly available thesis in advertising her/his skills and experience.

As part of the new archiving process, each student has to guarantee that she/he has the necessary rights and permissions to the content included in the thesis. Specifically, a student needs to ask a permissions for each figure and table not created by the student her/himself but copied from another source.

All theses in the University of Oulu that are to be approved are analyzed with a program called Turnitin. This program checks the amount of similar text in the thesis and already published documents (i.e., theses delivered earlier and documents in various databases and in the Internet). The role of Turnitin is twofold: First, it checks whether a student has copied text from others' documents. Copying of few sentences is acceptable when the copied text is referenced and quoted properly; otherwise such copying is not permitted and causes official consequences—the thesis can even be rejected. Second, when the thesis is in Turnitin's databases, it checks whether others have copied text from this thesis.

The student can choose whether their thesis is stored as comparison material in Turnitin. We recommend doing this, as then the thesis is protected against incorrect copying by others. However, once delivered to Turnitin the thesis cannot be deleted from its databases. Text copied from others' documents must be referenced and quoted properly in the first version delivered to Turnitin. Hence, a student must **always** check correct referencing and quoting with the supervisor before delivering the work to Turnitin! ITEE professors have additional tools for performing this check. The rights and permissions for thesis content can be checked at the same time.

The University of Oulu offers a free tool, Turnitin Draft Coach, to all students as part of Microsoft Office 365 Word Online. With this tool, you will receive feedback and can check the originality of your text and the correctness of the references already during the writing process. More information about the tool and its activation on the website of the University of Oulu at `https://www.oulu.fi/en/news/want-improve-your-academic-writing-and-achieve-better-results`.

Students can utilize the Turnitin similarity checking independently. The tool can be found in Moodle `https://moodle.oulu.fi/enrol/index.php?id=31794` (Link valid for 2026). The thesis is checked individually, so the originality of the text can be confirmed.

More information can be found from the degree programme web pages [4] and from Laturi's web pages: `http://laturi.oulu.fi`.

3.6. Optional Bound Versions

It is not mandatory to bind the thesis at a printing house. However, a student is free to bind the thesis when she/he wants to have a high-quality copy for relatives, for example.

For bound theses, select the *print* option in the thesis template, which adjusts the page margins to the following: left 4,5 cm, right 2,0 cm, top 2,5 cm, and bottom 3,0 cm. Verify that figures and tables are correctly positioned and that page numbers and page number references in the text are accurate, as these may differ from the online PDF version.

To ensure a high-quality thesis that is consistent with previous theses, the following rules apply: The words MASTER'S THESIS should be printed in the center of the front cover. The NAME OF THE AUTHOR should be printed in the lower right-hand corner of the front cover. The NAME OF THE AUTHOR and the year of publication should be printed on the spine of the thesis. Your thesis must be bound at a printing house specialized in printing theses. The printing house will also produce and attach the cover to the printout that you provide. The cover must be black, and the text must be printed directly on it; stickers are not allowed.

3.7. Copyright, Licenses and the Use of Images

Intellectual property rights (IPR) protect works, i.e., the authors and rights-holders of literary or artistic works that exceed the threshold of originality. These works include, for instance, texts, images, music, and computer software code. This section explains the related issues, especially concerning images, and provides guidelines for dealing with copyright matters in your thesis.

Your thesis is also a copyrighted work. If no license is explicitly stated, the license is "All rights reserved," meaning it can only be used in limited ways under copyright law. The Federation of Finnish Open Science Coordination in Finland and the University of Oulu recommend publishing theses as open access and under a CC-BY 4.0 license, or if warranted, another CC license [12]. CC-BY 4.0 allows reuse, including commercial use, of your work as long as the author is credited.

Among the various types of works protected by copyright, images present particular challenges and considerations. Images are frequently used to illustrate concepts, present data, or enhance the visual appeal of academic texts. However, their use is subject to specific copyright restrictions, and improper use can lead to legal and ethical issues. For this reason, figures found on the Internet, for example, may not be freely used in theses without proper permission or licensing.

According to the copyright enactments, you must always have permission from the copyright holder, which is not necessarily the author, to use their figure in your work. To summarize: **Copyright infringement may lead to legal actions and penalties, and in those cases the author is responsible.** The writer should grow towards to mainly using figures of his own in the thesis. Often it is necessary to redraw the figure and cite the original source.

It is good practice to explicitly state the copyright owners and license for all images in your thesis. This can be done via a blanket statement on a separate copyright page that explicitly mentions images, or for every image separately in the caption. All exceptions to the main license of your thesis (e.g., "All rights reserved" or "CC-BY 4.0") should **always** be included in the image captions.

If your use of external images is extensive and is a central part of your thesis, it is good to also explicitly list them on the copyright page or in an appendix, e.g., "Figure 5 is reproduced from [x] under the provisions of copyright law allowing use with appropriate citation." and also discuss their use with your thesis supervisor at an early stage and acquire the necessary permissions.

The starting point is that the figures must be related to the text, i.e, the text must explain why the figure has been presented. The text must contain a reference to the figure. Use self-made figures (drawings, schematics, simulation or measurement result figures, photographs), if possible.

It is a good practice to mark the photographer in the photos (e.g. "Photographer first name last name year.") If you are not the photographer, ask permission and keep the permission (e.g., email correspondence) safe. In addition, keep any other licenses you have received, including those under the CC license, safe. Often publications under a CC license mention the CC license on the front page and the figure is on page z. Save at least the page explaining the CC license from the publication from which you have taken the photo. For this reason, we recommend mentioning the license explicitly for your own significant contributions, even if it is the same as the main license.

As mentioned, in case the figure has been taken from elsewhere, the author needs to have a permission from the copyright holder. Usually this is based on a license that gives permission to use the work with some conditions. The most common example would be a CC-BY license, such as the one used in Figure 3. If the figure is from a publication you are citing, "Under/Reprinted/Adapted CC BY 4.0 license picture from [x] ©20xx Author." would be appropriate. In Finnish, use "[Muokattu] CC BY 4.0 lisenssin alainen kuva lähteestä [x]. ©Vuosi Tekijä."



Figure 3: "Stephen Crane, Author, Red Badge of Courage" by Tony Fischer Photography is licensed under CC BY 2.0.

The permission can also come from the publisher or, if applicable, your employer (in cases where they hold the copyright). For instance, IEEE and many other publishers have an automated system for granting permission for theses. Mark the permission in the caption as requested by the right holder, e.g., "©2022 IEEE. Reprinted, with permission, from [x]."

Not all figures exceed the threshold of originality. These include, for example, scientific diagrams and flowcharts. Good research practices require that even in this case, the source is

cited in the caption ("Figure from [x]."). However, it would be best to ask for permission or at least redraw the image. Images with a CC license are free to use, so long as the terms of use are followed [13]. Even then, the caption must mention the source (article, website, etc.) and that the image is CC-licensed.

Based on the right to quote, it is possible to take a photocopy, screenshot or photograph of a work (not the original file) and publish it, citing the original source in the caption. A critical aspect of the right to quote is that the use of the image is relevant to the work, i.e., the image must be related to the subject, and it must be explained in the text (this applies to all images). Similarly, the source must be cited in text citations.

Sometimes you may want to use someone else's original image idea as a basis for your own work. Permission is not required provided that your work clearly an amply edited version of the original. However, in accordance with good research practices the type of modification and source ("Modified/Adapted/Redrawn/Based on/Created using data from [x]") and must be declared in the caption. Should you only make minor edits to the original, separate permission is required. The University of Oulu Acta Oulu publishing guide [14] has more details on this topic, and also the proper wordings in Finnish [15].

When searching for images through an internet image search, it is not always easy to find the rights holder from whom to request permission for use. Hunting for a permit can be a demanding task. The easiest way is to use images and figures from scientific articles (or with open licenses) as in these the rights holders are clearly declared.

3.8. Accessibility

Put effort into the accessibility of your thesis. The structure of an accessible document is clear, and the wording is as understandable as possible. Figures and tables should be given an alternative text that summarises the information they contain for the reader. Alternative text is especially important in situations where the reader cannot see the figure or table, for example due to vision impairment. More information on accessible Word documents and archivable PDF files can be found in the texts of the University of Oulu Campus ICT [16, 17]. You can also read about the accessibility of mathematical equations in the Campus ICT text [18].

4. THE LITERARY STRUCTURE OF A THESIS

The structure of a text is based on a pre-designed content structure (table of contents and the body) that can vary a great deal from thesis to thesis depending on the topic as well as the scope of the thesis. The presentation order of the first few pages is fixed, and should be presented as it is described in Sections 4.1 to 4.7 in these instructions. When applying page numbering, the title page has page number '1', but page numbers are shown (at the right upper corner of each page, Arabic numerals) only from Chapter 1, Introduction, onwards.

4.1. First Pages

The first pages include the title page, abstract(s), table of contents, foreword and list of abbreviations and symbols.

4.1.1. Title Page

In case the title page template of these instructions can't be used, you can produce the title page yourself following the model available from the degree program web pages. Please ensure that there are no typos, and that your degree program is stated correctly. Your name should be written into the middle. The title should be written below your name with capital letters, centered, and divided into several lines as necessary to produce a balanced appearance. University logo and the faculty name are placed at the top and the type of the thesis, degree programme name, and date are placed at the bottom.

4.1.2. Abstract

The abstract of your thesis will be fed into various databases and catalogues. It should crystallize the essence of your thesis. A good abstract is a bait that attracts the reader to take a closer look of the content. The abstract should be self-contained, i.e., the reader should be able to get a clear picture of your thesis from the abstract alone. There must not be any references to your thesis or other sources, but it should also not include any information not found in your thesis. The abstract should include the main elements, as well as the methods used and results obtained, and main conclusions of your thesis. The recommended length of an abstract is 200 words. Rare terminology and abbreviations should be avoided.

The bibliographic information of the thesis should be printed at the top of the abstract page. The keywords of your thesis should be printed below the abstract. The recommended number of keywords is 2–6 keywords or word sets. It is recommended that the keywords are not any words included in the title of the thesis. Keywords serve an important purpose for anyone performing literature searches in a library and other information catalogues. An example abstract is presented in Appendix 1.

4.1.3. Abstract in Finnish

You also have to write your thesis abstract in Finnish, or have your abstract translated into Finnish. It should be written in flawless Finnish. If the language of the thesis is English, the abstract in English is placed first, and abstract in Finnish (Tiivistelmä) after it. If the language of the thesis is Finnish the order is opposite. The requirement of the Finnish abstract also apply to international students.

4.1.4. Table of Contents

The table of contents lists the chapters with their headings and subheadings and their respective page numbers. However, the page number is shown in the table of contents only from the first chapter, Introduction, onwards.

4.1.5. Foreword

The foreword page should describe the aim of the thesis, and its various research stages, and present the partners, funding and circumstances involved in the thesis project. The forewords should also include words of gratitude, addressed to people who have been incremental in your thesis-writing process. The supervisor, the second examiner, and the technical supervisor can be mentioned as well.

4.1.6. List of Abbreviations and Symbols

The abbreviations of terms, variables and measurement units as well as the definitions of other symbols are given as a single list in alphabetical order stylized exactly as in the body text (see Section 4.3) on this page. You should check the validity of all abbreviations and symbols from standards and other reliable sources. Concerning measurement units, you should apply the internationally approved SI-system of symbols [19, 20] and quantities and units specified in the IEC 80000-13 standard. For example, 1000 bytes is one kilobyte (1 kB, 'B' is Bytes, 'b' is bits) and 1024 bytes is one kibibyte, 1 KiB. Similarly, 1 048 576 bytes mebi (Mi); the larger units are gibi (Gi), tebi (Ti), pebi (Pi), and exbi (Ei).

In the list, the abbreviations are defined first, then mathematical and other symbols and last letter symbols so that Latin, Greek and other letters are each presented in their designated groups.

4.2. Introduction

In the introduction, you should describe the background and motivation of your thesis, introduce the reader to your research questions and methodology, describe in detail the objectives of your thesis (i.e. what is your aim to achieve), and on what basis the scope of your research area has been chosen. You can also state explicitly what will not be covered. You should cite earlier work done in the field. You should **not** discuss the results of your thesis in the introduction.

You can start by introducing some need of an individual user or the society, or some well-known application or technology. Specifying your own topic as part of such larger entity helps the reader to understand the work and estimate its significance and success. Note that you should avoid specifying too challenging objectives, as failing to meet the objectives can reduce the grade.

Nowadays master's theses in engineering are often a part of wider research projects at universities or industries, and it might therefore be difficult for the reader to discern when the author is describing her/his personal work, and when is she/he describing the work of a research group. In cases like these, the author should describe as well as possible what exactly was her/his role and contribution in the research/project. At the end of the introduction, you might want to give an overview about the structure of your thesis.

4.3. Core Text Part of Your Thesis

How you handle the core topic of your thesis depends essentially on the nature of your research/project. Most theses first describe the technological and scientific environment of the thesis and state-of-the-art related to the topic. This part should focus on presenting context for the author's own work and for the decisions made by the author, so that it is later clear why one approach, technology, or method was chosen over another.

When the author develops a part for a larger system, it is necessary to describe this larger system as well. For example, when an author develops a software component for an LTE base station, the base station and LTE telecommunication network need to be described. However, the author needs to pay attention to describe the larger system only at a reasonable level of detail. The rule of thumb is that detailed information should be given only when this information is necessary to understand the rest of the thesis.

In the beginning, you can describe and analyze optional approaches or methods, e.g., by applying system-level modeling. As stated above, you can also present solutions proposed in literature sources. The best grades require presenting the state of the art of both technology and science—that is, presenting both the already available solutions and the ongoing research. Such presentation shows that the author possesses good knowledge on the thesis' topic. Hence, it is crucial to get familiar with the latest literature on the thesis's topic as soon as possible. Books, scientific articles, and patents can be searched from the sources offered by the university library. The library's subject guide on information technology[7] is a good starting point for searching literature.

A theoretical approach on the given topic can become a natural element in some theses where the student presents, based on his on her own prior knowledge or referring to literature, the theoretical grounds on which the thesis work relies. However, unnecessary wordiness must be avoided and as a result the theory presented must be directly linked to the realised work. It is noteworthy, that many theses do not include a theoretical section of any type. **Hence, you should not insist on adding theory if it does not fit in naturally.**

In the possible presentation of related theory, equations and mathematical notations often play a prominent role. However, it is important to keep in mind that mathematics is a tool for technical writing – not the main purpose. Therefore, introducing all details with a mathematical equation is unnecessary. It is sufficient to present the basic equations, necessary variables and

[7] http://libguides.oulu.fi/c.php?g=58694

the end results. If needed, the full derivation can be added to the thesis as an appendix. In science and technology, two types of equations are employed:

- equations between quantities where letter symbols represent physical quantities, and
- equations between numerical values where letter symbols represent the numerical values of the variables.

A quantity is the product of the numerical value and the unit of measure. The unit of measure must always be separated from the preceding numerical value with a spac (e.g., 5 Gbit/s, 5 °C, yet 5°). Equations between quantities are recommended because they are not dependent on the choice of units of measure, unlike equations between numerical values. Calculations with quantities follow the rules of algebra, and the utilized symbols are typically single letters. The mathematical variables and symbols of quantities must be italicized, i.e., written in italics (e.g., area A, electric charge q). Vectors must be italicized and boldfaced (e.g., acceleration $\mathbf{a}$).

Numbers, functions, chemical symbols, units of measure, and subscripts are not italicized (e.g., 200, cos, log, NaCl, μA, cm, electron mass m_e), but subscripts representing symbols of quantities are italicized (e.g., thermal velocity v_T). The lowercase Greek letters used as symbols of quantities are italicized, but uppercase letters are not italicized (e.g., δ vs. Δ). More instructions for writing symbols can be found in the SI Guide [21], and some of them may be discipline-specific. List the symbols in the list at the beginning of your thesis, stylized the same as they appear in text and equations. Each equation must be part of a complete sentence. A blank line is left above and below an equation, and the equations are numbered in an increasing order through the entire text or, alternatively, by chapter if there are large quantities of equations. The corresponding order number is placed in parentheses on the right side of the equation. In the text, the equations are referred to by the order number of the equation in parentheses. In steady movement, speed v is

$$v = \frac{s}{t}, \tag{1}$$

where s is the distance and t the time travelled. The implications of equation Equation (1), in turn, are. . .

After the opening sections of your thesis, you should present the actual personal contribution your thesis is making, although some of this can already be included in the previous chapters. Describing the phases of the development work provides a clear structure, that is, describing first the requirements for the solution, then the design and implementation, followed by testing (i.e. verification).

The requirements can be either given by the customer or the author might need to determine the requirements her/himself by interviewing experts, for example. In either case, the actual work must not be straightforward following of instructions given by others but the author must provide a significant contribution to the solutions. Thesis work is independent, analytical problem solving and engineering work, in which the author weighs different solutions for the encountered problems and presents the reasoning and justifications for the selections she/he make during this process. Essential selections can be presented, for example, in a table listing the alternatives and selection criteria, and giving justified points for the alternatives for each criterion. For example, when a certain device needs to be selected, the criteria can be price, performance, and reliability, and each candidate devices is given points for each criterion.

The design should be presented using the established tools and representations. For example, UML diagrams can be used to describe software design. The aim of testing is to verify the

solution. When the requirements are presented in an earlier chapter, it is natural to test the solution against the requirements. So that your observations will benefit your readers, you should record and publish them in your thesis in as thorough and detailed a way as possible in their original format (such as tables, etc.). You must be careful not to mix actually obtained results and your estimates of the results. The reliability of the results can be analyzed using statistical methods, though strong conclusions require large samples.

In construction, or software-based theses, you should approach your research questions by means of system planning. You should only describe the necessary details of your basic theory and construction, which are directly relevant to the construction or software. You can include a broader description of the constructions and/or software in an appendix if you deem it important. You should describe the details of your construction/software in sections, starting from a general level, and then going into more detail, level by level. You should avoid a too detailed description concerning the operational principles of software structures and electronic circuits. However, details that are essential to the thesis—especially details that cover new ground or are not clear even to experts in the field—should be included in the thesis. Measurements are intrinsically related to theses which present constructions and therefore they must be carefully planned. The same is valid for the testing of software solutions. Yet, the thesis is not a measurement report and hence it is unnecessary to introduce all the results. Consequently, each presented figure must have a clear significance.

These instructions give guidelines on writing the core text of your thesis when your topic is to develop a concrete solution that can be tested, like a device, piece of software, or an algorithm. Even if the work does not produce such a concrete result, the analytical problem solving has to be present in the documentation and the result be presented clearly. For example, for a literature study, presenting the subject matter and the essential content found from the literature is not enough, but the author has to classify this content or otherwise analyze it. In topics not producing concrete results, such analysis is a mandatory requirement for the best grades. Specifically, it is always crucial to analyze how well the results meet the objectives set in the introduction. This analysis can be placed in the discussion chapter.

4.4. Discussion

A good thesis or other scientific work always has a discussion. In order to write this, you should be able to look at your work as if from a distance, to "step out of the box", as they say. You should be self-critical, compare your work to similar work in the field, and think analytically. You should be able to crystallize the results of your work, and put them into words. This can often be difficult even for an experienced writer, but it helps if you are well acquainted with published literature in the field.

A justified analysis of meeting the thesis' objectives is an important part of the discussion, and generally analysis of the results. When the requirements for the developed solution are presented earlier in the thesis, it is natural to first discuss how these requirement were fulfilled and from that conclude how well the objectives were met. Own solution should also be compared to the state of the art (technology and science) presented at the beginning. Your solution does not have to be better, but this kind of analysis is essential in engineering work and hence an important part of master thesis as well. Such a comparison is needed for the best grades.

The analysis can contain also more general commenting, for example, what was easy and what was difficult in the work. You can also discuss the overall significance of your thesis on a

more general level. Outlining potential further development based on your thesis, is valuable, especially if you have put forth clearly new or groundbreaking ideas. However, you should avoid unnecessary speculation here, as well as elsewhere: all statements should be well reasoned and brief. Other requirements for discussion can be found from the supervisors' evaluation instructions that are available at the degree programme web pages [4].

4.5. Summary

In the summary (or conclusions) you should present clearly in a nutshell the objectives of your thesis, its main content, your results, and the significance of your results. You should lay special emphasis on your results if you feel you have accomplished something. In summary you should not make references, or present any results not found elsewhere in your thesis.

The abstract and the summary overlap to a certain extent; they both describe the main contents and results of a thesis. However, the nature of the summary is broader and it does not have to be self-contained. In it, you should describe your aims, and you could describe any optional solutions or approaches, and motive the choices you have made. The abstract on the other hand could just describe in detail the solutions and approaches chosen for the thesis, leaving the optional approaches out.

4.6. References

The use of references serves many purposes. Scientific method relies on familiarizing with the topic and state of the art. However, efficient referencing can also compress your text, as you can leave the details in the references and repeat only the most important results.

The literature survey should be close to exhaustive, and this means that most of the information you present is taken from references. If a piece of information is not derived or devised by you, it is borrowed, and the origin of the information must be stated. Presenting somebody else's finding as your own is a scientific theft (plagiarism) that has serious consequences.

You should refer to original sources of the data—for example, to a book and not the handouts made based on the book. Be careful when referencing: the things you state really need to be found from the reference.

The references are mostly cited in your own words, and direct quoting is used only if you want to emphasis the source. In this case you place the quote in quotation marks, for example saying that the original phrasing of Moore's law is "The complexity for minimum component costs has increased at a rate of roughly a factor of two per year (see graph on next page)." [22]. Longer quotations (several lines) are placed in an indented paragraph (omit quotation marks). As Moore stated:

> Certainly over the short term this rate can be expected to continue, if not to increase. Over the longer term, the rate of increase is a bit more uncertain, although there is no reason to believe it will not remain nearly constant for at least 10 years. [22]

Correct referencing is important as the archiving of electronic theses makes them easily accessible (either in the library or in the Internet, depending on the publicity level selected by the thesis author). The reader must be able to easily find the original source to both assess your work and for the authors of the reference to be properly credited.

The sources are cited and the bibliography is compiled according to the IEEE style [23]. Sources are numbered in increasing order and listed in the order in which they appear in the text. References in the text are then indicated with a reference number, e.g., [1] or [1, 2, 8], which is placed inside square brackets, inside sentence punctuation with a (non-breaking) space before the bracket. References to books should also include the page number, for example [1, p. 15] or [1, pp. 15–17]. Figures and equations can be cited using: [1, Fig. 3]; [1, eq. (3)]. When writing in Finnish, you should adapt the IEEE guidelines, e.g., [1, s. 15];[1, s. 15–17]; [1, Kuva 3]; [1, Yhtälö (3)].

The reference is placed immediately where it is used or at the end of the sentence it relates to. Grammatically it can treated as a noun or a footnote, e.g., "As shown by Lappalainen [1]" or "As shown in [1]". The authors are only named if they are relevant for the sentence. **If no citation is mentioned, the sentence is assumed to be your own idea**.

In IEEE style, if multiple sentences refer to the same source, the citation should normally be repeated for each sentence to avoid ambiguity. If it is made explicitly clear that the same source continues – for example, by using linking phrases ("According to Smith [5]. . . He states. . . ") — then repeating the citation for each sentence is not strictly necessary.

If an entire paragraph is paraphrased from one source, you may place a single citation after the final sentence (as demonstrated here by placing [1] at the very end of this paragraph). This approach should be used sparingly in scientific writing, but it can be acceptable in a thesis literature review where a subsection is devoted to a single source. In all cases, ensure that paraphrased material is written entirely in your own words. Avoid covering long page ranges from books or articles in one block – instead, summarize in smaller, focused chunks with exact citations in each sentence. [1]

Referencing is easiest to do using LaTeX for writing the thesis and following the model given in these instructions. You should add your own references to the file *citations.bib*. A bibliography manager, such as Zotero [24, 25], is strongly recommended, as it often automatically fills in the required fields of the references.

References need to be presented so that it is clear what information is from a source and what is created by the author. Also the sources of equations, figures, and tables need to be given, when not created by the author her/himself. In addition, permissions need to be asked for tables and figures. Special attention needs to be paid to web sources, as web content can have a short life cycle. Hence, the date of downloading referenced web content should always be mentioned in the reference.

The examples below illustrate correct references. All authors should be listed, the names of journals and series should be written completely without abbreviations, and only the first letter of the title and proper nouns in the title should have uppercase letters. The DOI (Digital Object Identifier) of the publication should be included in the reference, when possible. This identifier locates the publication in an unambiguous fashion and hence facilitates the work of a reader searching the reference. The DOI can be found from the publications front page or from publishers' databases (e.g. IEEE Xplore, Springer Link, ACM Digital Library). When the reference has a DOI, you do not have to include an URL for it.

The *citations.bib* file in Overleaf ITEE MSc thesis template should guide you to use correct citation formats. The initials of the author's first names are written first, followed by the last name. If you are using other text preparation tools such as Microsoft Word, you should cite articles in journals as [26], in series as [27], and in books as [28, p. 55]. A section of compiled work should be cited as [29], and articles in conference proceedings as [30]. The citation to a thesis should look like [31], and patents should be cited in the manner of [32]. In some cases it

is necessary to cite a source in the Internet [8], and publications that lack a known author [33]. Please, see the References chapter for examples.

4.7. Appendices

Things you can include as an appendix are, e.g., derivations of equations or formulas, details of important computer programs, various tables, or performance characteristics and descriptions of special equipment or components applied in the thesis work. You can also include construction drawings and parts of catalogues in the appendix. Appendices are titled as shown before. As with figures and tables, all appendices you include should have a clear meaning—the number of appendices itself is not a merit.

If your text seems to contain a lot of references to an appendix, it may be easier for the reader that you copy the information (e.g., a schematic) into the text. This way the reader does not need to browse between pages.

Large block diagrams or schematics can be copied on A3 size paper. This needs to have two bends (Z-like) so that the right side of the appendix lies top and with a width of ca 2/3 of A4 page.

5. SUMMARY

These instructions describe the various stages of writing a master's thesis. We have presented the role of a Master's thesis in an engineering degree, and we have discussed about the importance of keeping close contact with your thesis supervisors. We have also described the procedure, according to which a master's thesis in degree programmes at ITEE, is to be written.

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7. APPENDICES

Appendix 1 Example Abstract

Note 1: Page numbering is done by consecutive numbering. Additional appendices, that are attached to the thesis (such as copies, drawings, etc.) are left without page numbers and placed at the end of the thesis.

Jurmu M. (2007) Resource Management in Smart Spaces Using Context-Based Leases. University of Oulu, Department of Electrical and Information Engineering. Master's Thesis, 77 p.

ABSTRACT

The convergence of wireless access networks in conjunction with the increased computing power of the handheld terminals is preparing the emergence of the ubiquitous computing paradigm. In the center of this paradigm are smart spaces, which are local environments saturated with various embedded computational resources. These spaces co-operate with mobile client devices in enabling advanced, service-oriented computation scenarios. This co-operation is typically enabled through the utilization of distributed and modular middleware frameworks. An emerging additional requirement however is the possibility to harness resources from the proximity environment to the mobile device in an on-demand fashion. This is a challenge especially to the resource management infrastructure of the smart spaces.

This thesis explores the concept of a smart space and presents a review of the current research and technologies. Subsequently, a lease-based design for resource management in smart spaces is presented. Leases in this work are negotiated agreements between the mobile clients and the resource management infrastructure, regarding the harnessed resources. Leasing is seen as a suitable solution for the management due to the transient nature of the resource usage. The inclusion of additional contextual features to the leases further facilitates the management.

Requirements for the design are derived from the review and from an example usage scenario. The requirements include dynamic mapping and contracting of resources from the proximity environment, monitoring of the contract validity, access control towards the resources and dynamic maintenance of the smart space infrastructure. Presented design is analytically compared against existing solutions, and several points for future development are listed. According to the comparison, none of the existing solutions utilize contracts with contextual validity in resource management. Two publications of this work have been accepted into an international conference and a workshop on the focus area of pervasive computing.

Keywords: Ubiquitous computing, mobile computing, task-based computing, context-awareness, QoS.